# WATER HEATER TEST STATION | PCB Wiring & Connection Guide

Embedded Hardware System for Control, Data Acquisition & AC Monitoring

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**[!] DANGER — HIGH VOLTAGE [!]** Do **NOT** open this enclosure unless all power sources are de-energized. **L1 and L2 (AC mains) are present on the PCB.** Valve L1 is wired **upstream of the breaker** and remains **energized even when the WH breaker is OFF.** Before opening: turn **OFF the main panel breaker**, disconnect the **24 VDC supply**, and unplug the **Raspberry Pi USB-C power**.

## Wire Colors Legend

Colored **boxes** indicate the signal type or function. Colored **arrows** represent the actual wire color used in the station wiring. Any wiring must follow this guide or be updated to match configuration changes.

- **AC Power**
  High voltage: L1 and L2 (240 VAC) or 120 VAC configuration
  Includes relay switching (valve control)
- **Sensor Signals**
  4–20 mA transmitter loops
- **Low Voltage**
  24 VDC and USB-C 5 VDC
- **Communication**
  RS-485, HDMI (display), LED indicators
- **Neutral (N)**
  120 VAC return (not used in 240 VAC mode)
  Used instead of L2 for 120 VAC water heater

**Flow Transmitter**
** WH3 & WH4 use alternate wire colors*

## Station Overview

| Unit | Panel | Bkr | AC Config |
|---|---|---|---|
| **WH1** | P1 | Pos.1 | 240V (L1–L2) |
| **WH2** | P1 | Pos.3 | 120V (L1–N) |
| **WH3** | P2 | Pos.2 | 240V (L1–L2) |
| **WH4** | P2 | Pos.4 | 240V (L1–L2) |

**Panel Breakers:** Each panel contains two breakers: Panel 1 → WH1 and WH2 Panel 2 → WH3 and WH4.

**Main Breakers:** FAB-NDP1P-A (1,3) for Panel 1 FAB-NDP1P-A (2,4) for Panel 2.

**Important:** Turning OFF panel breakers does **NOT** de-energize the valves. All Valves remains live.

**Required for Safe Work:** Disconnect panel supply cables for both panels **OR** turn OFF the main breakers before any wiring changes.

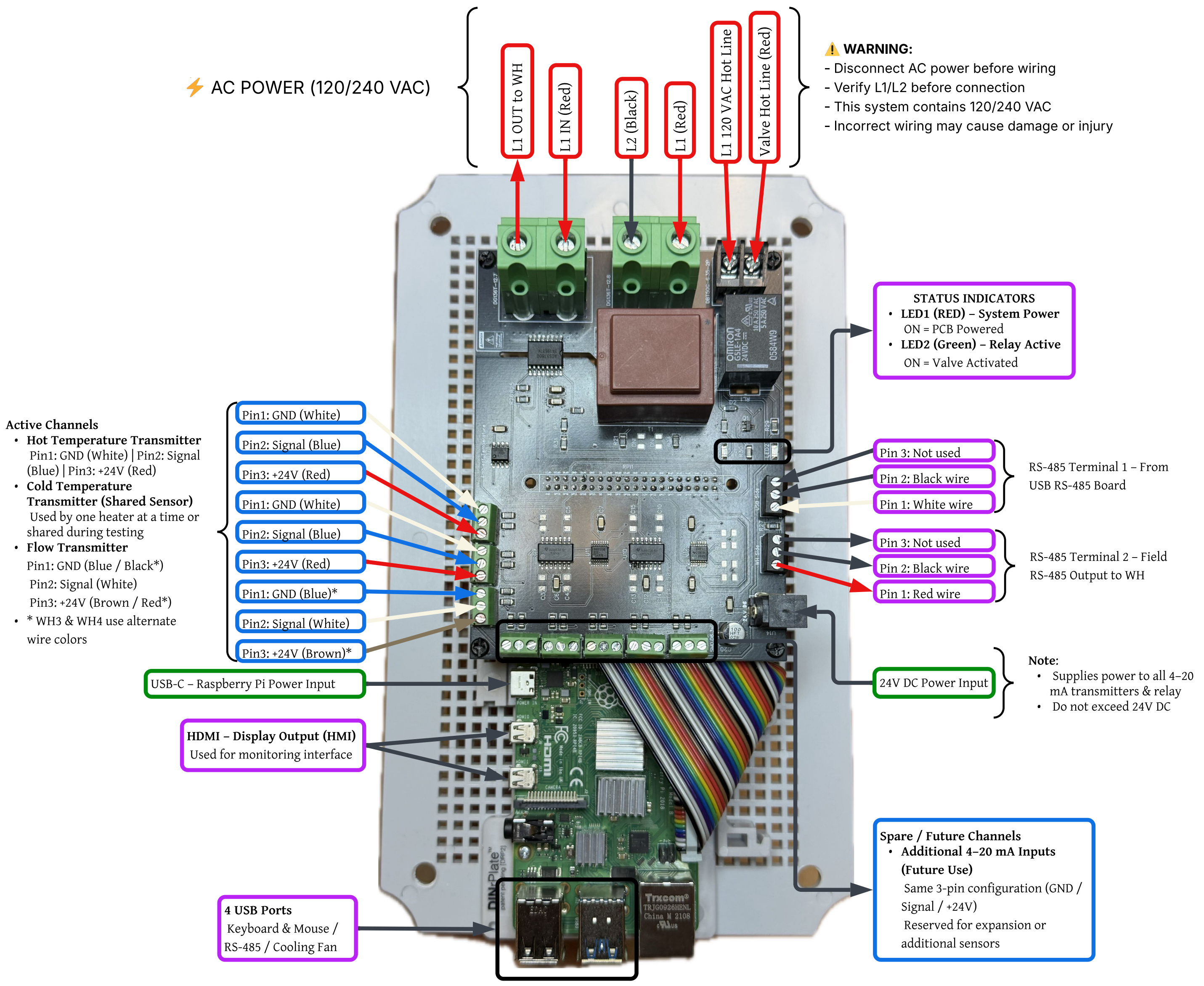
**Pipe Order:** Temperature Sensor (Hot) → 120 VAC Valve → Manual Valve → Flow Transmitter → Sink.

**Voltage Configuration:** System designed for 240 VAC (L1 + L2).
WH2 currently uses 120 VAC (Neutral instead of L2).
*Verify wiring before any configuration change.*

## Status LEDs

| LED | Colour | St. | Meaning |
|---|---|---|---|
| **LED1** | **RED** | ON | PCB + 4–20 mA ON |
| **LED1** | **RED** | OFF | No PCB power |
| **LED2** | **GREEN** | ON | Valve OPEN |
| **LED2** | **GREEN** | OFF | Valve CLOSED |

## PCB Connection Diagram (Annotated) — All connectors, pins, and wire colors as installed




## Quick Access & Notes

**Scan QR code for system documentation and updates**

- Full documentation
- Schematics & PCB design
- Software / Raspberry Pi setup
- Troubleshooting guide

## Connector Quick Reference

### 4–20 mA Transmitter Terminals

Pin 1 = GND | Pin 2 = Signal | Pin 3 = +24V

| Channel | P1 (GND) | P2 (Signal) | P3 (+24V) |
|---|---|---|---|
| Hot Temp. Tx | White | Blue | Red |
| Cold Temp. Tx (shared) | White | Blue | Red |
| Flow Tx WH1/WH2 | Blue | White | Brown |
| Flow Tx WH3/WH4* | Black | White | Red |
| Spare (5 ch.) | — | — | — |

**WH3/WH4 use alternate colours only for flow Tx — verify by pin number.*

### AC Power Terminals — LIVE / DE-ENERGIZE FIRST

| Terminal | Wire | Note |
|---|---|---|
| L1 IN | Red | From panel breaker |
| L1 OUT to WH | Red | To water heater (via current sensing) |
| L2 | Black | From panel breaker (240 VAC return) |
| N (WH2 only) | Yellow | Used only for 120 VAC water heater |
| Valve Hot (120V) | Red | LIVE even if panel breakers are OFF |
| Ground (PE) | Green | Protective earth (always connected) |

### RS-485 Terminals

| Terminal | Pin | Wire | Note |
|---|---|---|---|
| Term. 1 (USB brd.) | 1 | White | A(+) from Pi |
| | 2 | Black | B(−) |
| Term. 2 (to WH) | 1 | Red | A(+) to WH field |
| | 2 | Black | B(−) |

*Pin 3 unused on both terminals*

### Low Voltage Connections

**Low Voltage Connections**

| Connector | Function |
|---|---|
| USB-C | Pi power input (5V adapter) |
| HDMI | Display / HMI monitor output |
| 4x USB | Keyboard, mouse, RS-485 adapter, fan |
| 40-pin GPIO | Must match PCB pin labeling exactly |
| 24V Barrel Jack | Sensor / PCB power (+24V, min 1.5 A) |

**Note:** 24 VDC is supplied through the barrel jack in the station. If not available, use an external 24 VDC adapter (minimum 1.5 A).
**Pi Power:** Use the Raspberry Pi adapter or a stable 5 V supply.

## Power-Down Procedure (Before Opening Enclosure)

| | | |
|---|---|---|
| 1 | **OFF: MAIN breaker** | **OR disconnect panel supply cables (REQUIRED)** |
| 2 | **Verify with meter** | Confirm 0V at all AC terminals |
| 3 | **OFF: WH breaker** | Additional isolation only |
| 4 | **Unplug 24V adapter** | LED1 should turn OFF |
| 5 | **Unplug Pi USB-C** | Disconnect power cable |
| 6 | **Do NOT open panel** | Requires qualified electrician |

### [!] Key Safety & Wiring Notes

- **Valve L1 is pre-breaker:** remains live unless the main breaker is OFF or panel supply is disconnected.
- L1, L2, and N (WH2) are routed through the PCB. Do not touch while energized.
- Cold water transmitter is **shared** — connect to only one PCB at a time.
- LM35 ambient sensor is PCB-mounted only (not part of the pipe station).
- WH3/WH4 cable colors different. Verify by **pin number**.
- 4–20 mA check: 0.48 V to 2.40 V across 120 Ω.
- **WH2 Note:** currently configured for 120 VAC (L1 + N). If changed to 240 VAC, replace N with L2 and update calibration.